

DDL, DML, DCL & TCL

SECTION 1 : DDL(DATA DEFINITION LANGUAGE)

QUESTION 1:- CREATE TABLE

```
amrutha@Amruthas-MacBook-Air ~ % mysql -u root
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 18
Server version: 9.7.1 Homebrew

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE bank_db;
ERROR 1049 (42000): Unknown database 'bank_db'
mysql> CREATE DATABASE bank_db;
Query OK, 1 row affected (0.011 sec)

mysql> USE bank_db;
Database changed
mysql> CREATE TABLE bank_transactions (
  -> txn_id INT PRIMARY KEY,
  -> customer_name VARCHAR(50),
  -> branch_name VARCHAR(50),
  -> transaction_type VARCHAR(20),
  -> amount DECIMAL(10,2),
  -> transaction_date DATE
  -> );
Query OK, 0 rows affected (0.022 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_bank_db |
+-----+
| bank_transactions |
+-----+
1 row in set (0.002 sec)

mysql> DESCRIBE bank_transactions;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| txn_id         | int           | NO   | PRI | NULL    |       |
| customer_name  | varchar(50)   | YES  |     | NULL    |       |
| branch_name    | varchar(50)   | YES  |     | NULL    |       |
| transaction_type | varchar(20)   | YES  |     | NULL    |       |
| amount         | decimal(10,2) | YES  |     | NULL    |       |
| transaction_date | date          | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.007 sec)

mysql> █
```

QUESTION 2:- ALTER TABLE AND COLUMN

```
amrutha@Amruthas-MacBook-Air ~ % mysql -u root
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 20
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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE bank_db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SHOW TABLES;
+-----+
| Tables_in_bank_db |
+-----+
| bank_transactions |
+-----+
1 row in set (0.001 sec)

mysql> DESCRIBE bank_transactions;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| txn_id         | int           | NO   | PRI | NULL    |       |
| customer_name  | varchar(50)   | YES  |     | NULL    |       |
| branch_name    | varchar(50)   | YES  |     | NULL    |       |
| transaction_type | varchar(20)   | YES  |     | NULL    |       |
| amount         | decimal(10,2) | YES  |     | NULL    |       |
| transaction_date | date          | YES  |     | NULL    |       |
| account_no     | varchar(20)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.002 sec)
```

QUESTION 3:- ALTER TABLE MODIFY COLUMN

```
amrutha@Amruthas-MacBook-Air ~ % mysql -u root
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 21
Server version: 9.7.1 Homebrew

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE bank_db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> DESCRIBE bank_transactions;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| txn_id         | int           | NO   | PRI | NULL    |       |
| customer_name  | varchar(50)   | YES  |     | NULL    |       |
| branch_name    | varchar(50)   | YES  |     | NULL    |       |
| transaction_type | varchar(20)   | YES  |     | NULL    |       |
| amount         | decimal(10,2) | YES  |     | NULL    |       |
| transaction_date | date          | YES  |     | NULL    |       |
| account_no     | varchar(20)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.002 sec)

mysql> ALTER TABLE bank_transactions
-> MODIFY customer_name VARCHAR(100);
Query OK, 0 rows affected (0.020 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> DESCRIBE bank_transactions;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| txn_id         | int           | NO   | PRI | NULL    |       |
| customer_name  | varchar(100)  | YES  |     | NULL    |       |
| branch_name    | varchar(50)   | YES  |     | NULL    |       |
| transaction_type | varchar(20)   | YES  |     | NULL    |       |
| amount         | decimal(10,2) | YES  |     | NULL    |       |
| transaction_date | date          | YES  |     | NULL    |       |
| account_no     | varchar(20)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.002 sec)
```

QUESTION 4:- RENAME TABLE

```
amrutha@Amruthas-MacBook-Air ~ % mysql -u root
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 22
Server version: 9.7.1 Homebrew

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE bank_db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SHOW TABLES;
+-----+
| Tables_in_bank_db |
+-----+
| bank_transactions |
+-----+
1 row in set (0.001 sec)

mysql> RENAME TABLE bank_transactions
-> TO customer_transactions;
Query OK, 0 rows affected (0.007 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_bank_db |
+-----+
| customer_transactions |
+-----+
1 row in set (0.002 sec)
```

QUESTION 5:- TRUNCATE TABLE

```
mysql> USE bank_db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SHOW TABLES;
+-----+
| Tables_in_bank_db |
+-----+
| customer_transactions |
+-----+
1 row in set (0.002 sec)

mysql> INSERT INTO customer_transactions
-> (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date, account_no)
-> VALUES
-> (1, 'Rahul', 'Hyderabad', 'Deposit', 10000, '2026-08-05', 'ACC101'),
-> (2, 'Priya', 'Warangal', 'Withdraw', 5000, '2026-08-05', 'ACC102');
Query OK, 2 rows affected (0.004 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM customer_transactions;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 1 | Rahul | Hyderabad | Deposit | 10000.00 | 2026-08-05 | ACC101 |
| 2 | Priya | Warangal | Withdraw | 5000.00 | 2026-08-05 | ACC102 |
+-----+-----+-----+-----+-----+-----+-----+
2 rows in set (0.001 sec)

mysql> TRUNCATE TABLE customer_transactions;
Query OK, 0 rows affected (0.010 sec)

mysql> SELECT * FROM customer_transactions;
Empty set (0.001 sec)

mysql> DESCRIBE customer_transactions;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| txn_id | int | NO | PRI | NULL | |
| customer_name | varchar(100) | YES | | NULL | |
| branch_name | varchar(50) | YES | | NULL | |
| transaction_type | varchar(20) | YES | | NULL | |
| amount | decimal(10,2) | YES | | NULL | |
| transaction_date | date | YES | | NULL | |
| account_no | varchar(20) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.002 sec)
```

QUESTION 6:- DROP TABLE

```
amrutha@Amruthas-MacBook-Air ~ % mysql -u root
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 27
Server version: 9.7.1 Homebrew

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE bank_db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SHOW TABLES;
+-----+
| Tables_in_bank_db |
+-----+
| bank_backup        |
| customer_transactions |
+-----+
2 rows in set (0.001 sec)

mysql> DROP TABLE bank_backup;
Query OK, 0 rows affected (0.008 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_bank_db |
+-----+
| customer_transactions |
+-----+
1 row in set (0.001 sec)
```

SECTION 2 : DML(DATA MANIPULATION LANGUAGE)

QUESTION 7:- INSERT SAMPLE DATA

```
mysql> SHOW TABLES;
+-----+
| Tables_in_bank_db |
+-----+
| customer_transactions |
+-----+
1 row in set (0.001 sec)

mysql> DESCRIBE bank_transactions;
ERROR 1146 (42S02): Table 'bank_db.bank_transactions' doesn't exist
mysql> DESCRIBE customer_transactions;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| txn_id         | int           | NO   | PRI | NULL    |      |
| customer_name  | varchar(100)  | YES  |     | NULL    |      |
| branch_name    | varchar(50)   | YES  |     | NULL    |      |
| transaction_type | varchar(20)   | YES  |     | NULL    |      |
| amount         | decimal(10,2) | YES  |     | NULL    |      |
| transaction_date | date          | YES  |     | NULL    |      |
| account_no     | varchar(20)   | YES  |     | NULL    |      |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.002 sec)

mysql> INSERT INTO customer_transactions
-> (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date, account_no)
-> VALUES
-> (101,'Ravi','Hyderabad','Deposit',5000,'2024-01-05','ACC101'),
-> (102,'Sita','Hyderabad','Withdrawal',2000,'2024-01-06','ACC102'),
-> (103,'Kiran','Vijayawada','Deposit',12000,'2024-01-08','ACC103'),
-> (104,'Anil','Vizag','Deposit',8000,'2024-01-10','ACC104'),
-> (105,'Priya','Hyderabad','Withdrawal',3500,'2024-01-11','ACC105'),
-> (106,'Ramesh','Vizag','Deposit',15000,'2024-01-12','ACC106'),
-> (107,'Keerthi','Vijayawada','Withdrawal',1000,'2024-01-13','ACC107'),
-> (108,'Rahul','Hyderabad','Deposit',9000,'2024-01-14','ACC108'),
-> (109,'Sneha','Vizag','Withdrawal',4000,'2024-01-15','ACC109'),
-> (110,'Madhu','Vijayawada','Deposit',11000,'2024-01-16','ACC110');
Query OK, 10 rows affected (0.003 sec)
Records: 10 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM customer_transactions;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 101    | Ravi          | Hyderabad  | Deposit          | 5000.00 | 2024-01-05      | ACC101     |
| 102    | Sita          | Hyderabad  | Withdrawal       | 2000.00 | 2024-01-06      | ACC102     |
| 103    | Kiran         | Vijayawada | Deposit          | 12000.00 | 2024-01-08      | ACC103     |
| 104    | Anil          | Vizag      | Deposit          | 8000.00 | 2024-01-10      | ACC104     |
| 105    | Priya         | Hyderabad  | Withdrawal       | 3500.00 | 2024-01-11      | ACC105     |
| 106    | Ramesh        | Vizag      | Deposit          | 15000.00 | 2024-01-12      | ACC106     |
| 107    | Keerthi       | Vijayawada | Withdrawal       | 1000.00 | 2024-01-13      | ACC107     |
| 108    | Rahul         | Hyderabad  | Deposit          | 9000.00 | 2024-01-14      | ACC108     |
| 109    | Sneha         | Vizag      | Withdrawal       | 4000.00 | 2024-01-15      | ACC109     |
| 110    | Madhu         | Vijayawada | Deposit          | 11000.00 | 2024-01-16      | ACC110     |
+-----+-----+-----+-----+-----+-----+-----+
10 rows in set (0.000 sec)
```

QUESTION 8:- UPDATE

```
amrutha@Amruthas-MacBook-Air ~ % mysql -u root
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Your MySQL connection id is 29
Server version: 9.7.1 Homebrew

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE bank_db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SELECT * FROM customer_transactions WHERE txn_id = 105;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 105 | Priya | Hyderabad | Withdrawal | 3500.00 | 2024-01-11 | ACC105 |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.001 sec)

mysql> UPDATE customer_transactions
-> SET amount = 5000
-> WHERE txn_id = 105;
Query OK, 1 row affected (0.002 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> SELECT * FROM customer_transactions WHERE txn_id = 105;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 105 | Priya | Hyderabad | Withdrawal | 5000.00 | 2024-01-11 | ACC105 |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.000 sec)
```

QUESTION 9:- DELETE

```
mysql> DELETE FROM customer_transactions
-> WHERE txn_id = 111;
Query OK, 1 row affected (0.013 sec)

mysql> SELECT * FROM customer_transactions WHERE txn_id = 111;
Empty set (0.000 sec)
```

QUESTION 10:- SELECT

```
mysql> SELECT *
-> FROM customer_transactions;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 101 | Ravi | Hyderabad | Deposit | 5000.00 | 2024-01-05 | ACC101 |
| 102 | Sita | Hyderabad | Withdrawal | 2000.00 | 2024-01-06 | ACC102 |
| 103 | Kiran | Vijayawada | Deposit | 12000.00 | 2024-01-08 | ACC103 |
| 104 | Anil | Vizag | Deposit | 8000.00 | 2024-01-10 | ACC104 |
| 105 | Priya | Hyderabad | Withdrawal | 5000.00 | 2024-01-11 | ACC105 |
| 106 | Ramesh | Vizag | Deposit | 15000.00 | 2024-01-12 | ACC106 |
| 107 | Keerthi | Vijayawada | Withdrawal | 1000.00 | 2024-01-13 | ACC107 |
| 108 | Rahul | Hyderabad | Deposit | 9000.00 | 2024-01-14 | ACC108 |
| 109 | Sneha | Vizag | Withdrawal | 4000.00 | 2024-01-15 | ACC109 |
| 110 | Madhu | Vijayawada | Deposit | 11000.00 | 2024-01-16 | ACC110 |
+-----+-----+-----+-----+-----+-----+-----+
10 rows in set (0.002 sec)
```

QUESTION 11:- SELECT WITH WHERE

```
mysql> SELECT *
-> FROM customer_transactions
-> WHERE transaction_type = 'Deposit';
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05	ACC101
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	ACC103
104	Anil	Vizag	Deposit	8000.00	2024-01-10	ACC104
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12	ACC106
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	ACC108
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	ACC110

6 rows in set (0.002 sec)

QUESTION 12:- SELECT WITH ORDER BY

```
mysql> SELECT *
-> FROM customer_transactions
-> ORDER BY amount DESC;
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12	ACC106
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	ACC103
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	ACC110
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	ACC108
104	Anil	Vizag	Deposit	8000.00	2024-01-10	ACC104
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05	ACC101
105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11	ACC105
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15	ACC109
102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06	ACC102
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13	ACC107

10 rows in set (0.001 sec)

SECTION 3: DCL(DATA CONTORL LANGUAGE)

QUESTION 13:- GRANT SELECT

```
mysql> CREATE USER 'bankuser'@'localhost'
-> IDENTIFIED BY 'bank123';
Query OK, 0 rows affected (0.018 sec)

mysql> GRANT SELECT, INSERT
-> ON bank_db.customer_transactions
-> TO 'bankuser'@'localhost';
Query OK, 0 rows affected (0.002 sec)

mysql> FLUSH PRIVILEGES;
Query OK, 0 rows affected, 1 warning (0.007 sec)

mysql> CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY 'audit123';
Query OK, 0 rows affected (0.005 sec)

mysql> GRANT SELECT
-> ON bank_db.customer_transactions
-> TO 'Auditor1'@'localhost';
Query OK, 0 rows affected (0.001 sec)

mysql> FLUSH PRIVILEGES;
Query OK, 0 rows affected, 1 warning (0.002 sec)

mysql> SHOW GRANTS FOR 'Auditor1'@'localhost';
```

Grants for Auditor1@localhost
GRANT USAGE ON *.* TO `Auditor1`@`localhost`
GRANT SELECT ON `bank_db`.`customer_transactions` TO `Auditor1`@`localhost`

2 rows in set (0.000 sec)

QUESTION 14:- GRANT ALL PRIVILEGES

```
mysql> CREATE USER 'BranchManager'@'localhost'
[ -> IDENTIFIED BY 'branch123';
Query OK, 0 rows affected (0.004 sec)

mysql> GRANT ALL PRIVILEGES
  -> ON bank_db.customer_transactions
[ -> TO 'BranchManager'@'localhost';
Query OK, 0 rows affected (0.001 sec)

mysql> FLUSH PRIVILEGES;
Query OK, 0 rows affected, 1 warning (0.002 sec)

mysql> SHOW GRANTS FOR 'BranchManager'@'localhost';
+-----+
| Grants for BranchManager@localhost |
+-----+
| GRANT USAGE ON *.* TO `BranchManager`@`localhost` |
| GRANT ALL PRIVILEGES ON `bank_db`.`customer_transactions` TO `BranchManager`@`localhost` |
+-----+
2 rows in set (0.000 sec)
```

QUESTION 15:- REVOKE SELECT

```
mysql> REVOKE SELECT
  -> ON bank_db.customer_transactions
[ -> FROM 'Auditor1'@'localhost';
Query OK, 0 rows affected (0.002 sec)

mysql> FLUSH PRIVILEGES;
Query OK, 0 rows affected, 1 warning (0.002 sec)

mysql> SHOW GRANTS FOR 'Auditor1'@'localhost';
+-----+
| Grants for Auditor1@localhost |
+-----+
| GRANT USAGE ON *.* TO `Auditor1`@`localhost` |
+-----+
1 row in set (0.000 sec)
```

QUESTION 16:- REVOKE ALL

```
mysql> REVOKE ALL PRIVILEGES
  -> ON bank_db.customer_transactions
[ -> FROM 'BranchManager'@'localhost';
Query OK, 0 rows affected (0.002 sec)

mysql> FLUSH PRIVILEGES;
Query OK, 0 rows affected, 1 warning (0.002 sec)

mysql> SHOW GRANTS FOR 'BranchManager'@'localhost';
+-----+
| Grants for BranchManager@localhost |
+-----+
| GRANT USAGE ON *.* TO `BranchManager`@`localhost` |
+-----+
1 row in set (0.000 sec)
```

SECTION 4: TCL(TRANSACTION CONTROL LANGUAGE)

QUESTION 17:- COMMIT

```
amrutha@Amruthas-MacBook-Air ~ % mysql -u root
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 35
Server version: 9.7.1 Homebrew

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE bank_db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SELECT @@autocommit;
+-----+
| @@autocommit |
+-----+
|             1 |
+-----+
1 row in set (0.001 sec)

mysql> SET autocommit = 0;
Query OK, 0 rows affected (0.002 sec)

mysql> SELECT * FROM customer_transactions
    -> WHERE txn_id = 101;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
|    101 | Ravi          | Hyderabad  | Deposit          | 5000.00 | 2024-01-05      | ACC101     |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.001 sec)

mysql> UPDATE customer_transactions
    -> SET amount = 6000
    -> WHERE txn_id = 101;
Query OK, 1 row affected (0.002 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> COMMIT;
Query OK, 0 rows affected (0.002 sec)

mysql> SELECT * FROM customer_transactions
    -> WHERE txn_id = 101;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
|    101 | Ravi          | Hyderabad  | Deposit          | 6000.00 | 2024-01-05      | ACC101     |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.000 sec)
```

QUESTION 18:- SAVEPOINT

```
mysql> SET autocommit = 0;
Query OK, 0 rows affected (0.000 sec)

mysql> SAVEPOINT Before_Update;
Query OK, 0 rows affected (0.001 sec)

mysql> UPDATE customer_transactions
    -> SET amount = 7000
    -> WHERE txn_id = 102;
Query OK, 1 row affected (0.002 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM customer_transactions
    -> WHERE txn_id = 102;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
|    102 | Sita          | Hyderabad  | Withdrawal       | 7000.00 | 2024-01-06      | ACC102     |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.001 sec)
```

QUESTION 19:- ROLLBACK TO SAVEPOINT

```
mysql> ROLLBACK TO SAVEPOINT Before_Update;
Query OK, 0 rows affected (0.000 sec)

mysql> SELECT * FROM customer_transactions
-> WHERE txn_id = 102;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 102 | Sita | Hyderabad | Withdrawal | 7000.00 | 2024-01-06 | ACC102 |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.000 sec)
```

QUESTION 20:- FULL TCL EXAMPLE

```
mysql>
mysql> UPDATE customer_transactions
-> SET amount = 2000
-> WHERE txn_id = 102;
Query OK, 1 row affected (0.000 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.001 sec)

mysql> SET autocommit = 0;
Query OK, 0 rows affected (0.000 sec)

mysql> UPDATE customer_transactions
-> SET amount = 7000
-> WHERE txn_id = 101;
Query OK, 1 row affected (0.001 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> SAVEPOINT SP1;
Query OK, 0 rows affected (0.000 sec)
```

```
mysql> UPDATE customer_transactions
-> SET amount = 9000
-> WHERE txn_id = 102;
Query OK, 1 row affected (0.000 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> ROLLBACK TO SAVEPOINT SP1;
Query OK, 0 rows affected (0.000 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.001 sec)

mysql> SELECT txn_id, customer_name, amount
-> FROM customer_transactions
-> WHERE txn_id IN (101,102);
+-----+-----+-----+
| txn_id | customer_name | amount |
+-----+-----+-----+
| 101 | Ravi | 7000.00 |
| 102 | Sita | 2000.00 |
+-----+-----+-----+
2 rows in set (0.000 sec)
```